HDE Note

1 Basic Knowledge & SVCDE Methods for ODE

1.1 Linear Algebra

Matrix Calculate: $det(kA) = k^n det(A)$ || $det(A^{-1}) = det(A)^{-1}$ || $(AB)^* = B^* A^*$ || $det(A) = det(A^T)$.

Eigenvalue λ & Eigenvector x: Def: $A\mathbf{x} = \lambda\mathbf{x}$. Method: $^1 det(A - \lambda I) = 0 \to \lambda$. $^2 (A - \lambda I)\mathbf{x} = 0 \to \mathbf{x}$.

· **Algebraic Multiplicity (AM) of λ**: λ 重复出现的次数. ≥ **Geometric Multiplicity (GM) of λ**: # **x** under λ.

· **Generalised Eigenvectors**: If AM>GM: Using $\xi_\lambda = (A - \lambda\mathbb{I})\eta_1, \eta_{n-1} = (A - \lambda\mathbb{I})\eta_n$ for each ξ_λ. η_j are *generalised eigenvectors*.

1.2 Calculus & SVCDE Method for ODE

Some Defs: $\nabla F = (\partial_x F, \partial_y F, \partial_z F, ...)$ || **Simply Connected**: 没有洞, 相互连接的区域. || ∂D: 区域 D 的边界.

Method of Integrating Factors: $y' + p(x)y = q(x)$. $\Rightarrow$ Integrating Factor: $\mu = e^{\int p(x)dx}$, $\mu\, y = \int q(x) \cdot \mu\, dx$

Homogeneous 2nd ODE form: $ay'' + by' + c = 0 \Rightarrow$ Calculate $ar^2 + br + c = 0$, ps: $\Delta = b^2 - 4ac$, roots: r_1, r_2

· 1. $\Delta > 0, y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$; 2. $\Delta = 0, y = (C_1 + C_2 x)e^{rx}$; 3. $\Delta < 0, y = e^{\alpha x}[C_1 cos(\beta x) + C_2 sin(\beta x)]$ ps: $r_1, r_2 = \alpha \pm \beta i$

non-Homogeneous 2nd ODE form: $ay'' + by' + c = g(x) \Rightarrow y = y_h + y_p$ ps: y_h is sol for $g(x) = 0$.

· Method of Undetermined Coefficients: Let $A_n(x) = \sum_{i=0}^n a_i x^i$, $B_n(x) = \sum_{i=0}^n b_i x^i$, $P_n(x) = \sum_{i=0}^n p_i x^i$.

1. If $g(x) = P_n(x) \Rightarrow y_p = x^s A_n(x)$, $s :=$ $0, (0 \neq r_1, r_2)$; $1, (0 = r_1\ or\ r_2)$; $2, (0 = r_1 = r_2)$.
2. If $g(x) = Me^{kx} \Rightarrow y_p = Cx^s e^{kx}$, $s :=$ $0, (k \neq r_1, r_2)$; $1, (k = r_1\ or\ r_2)$; $2, (k = r_1 = r_2)$.
3. If $g(x) = Mcos(kx) + Nsin(kx) \Rightarrow y_p = x^s(C_1 cos(kx) + C_2 sin(kx))$, $s :=$ $0, (r_1, r_2 \neq ik)$; $1, (r_1, r_2 = ik)$.
4. If $g(x)$ = product from 1,2,3. $\Rightarrow y_p$ = corresponding product of y_p (x^s 只乘一次) , $s :=$ 令 y_p 与 y_c 没有相同项. (≤ 2)

1.3 ODE / PDE Definitions

Def of DEs / ODEs / PDEs: Given independent variables ($x_1, ..., x_n$), functions ($y_1(x_{j_1}), ..., y_p(x_{j_p})$) and some derivatives (e.g. $\partial_s y_t$).

· **DEs**: $\mathcal{F}_i[y_r, ..., \partial_{x_t} y_b, x_1, ..., x_n] = 0$, $i = s, s-1, ..., 1$. || **ODEs**: $n = 1$ (无偏导) **PDEs**: $n > 1$ (有偏导)

Def of Linear: Let L 去掉 $y_i^{(n)}$ 的函数 (e.g. $L = \frac{d^2}{dx^2} - x$ for $y'' - xy = x^2$) $\Rightarrow$ Linear if: $L[a_1 y_1 + a_2 y_2] = a_1 L[y_1] + a_2 L[y_2]$

1.4 (ODE) Uniqueness Theorem & Autonomous

Existence & Uniqueness Theorem (Real func): Given $\mathbf{y}'(t) = \mathbf{F}(t, \mathbf{y}^T)$, with $\mathbf{y}(t_0) = \mathbf{y_0}$. Let $R_i = [t_0 - \delta, t_0 + \delta] \times \prod_j [y_{0_j} - \delta_j, y_{0_j} + \delta_j]$.

· If F_j , $\partial_{y_i} F_j : R_i \to \mathbb{R}$ are continuous. $\Rightarrow$ $\exists$ unique solution **y** in R_i' Let $R_i' = [t_0 - \varepsilon, t_0 + \varepsilon] \times \prod_j [y_{0_j} - \varepsilon_j, y_{0_j} + \varepsilon_j] \subseteq R_i$.

Autonomous: For $\mathbf{y}'(t) = \mathbf{f}(t, \mathbf{y}^T)$. If $\partial_x \mathbf{f} = \mathbf{0} \Rightarrow$ autonomous. or: $\mathbf{f}(t, \mathbf{y}^T) = \mathbf{f}(\mathbf{y}^T)$ (不含 t)

2 Linear ODEs

n-th ODE → 1st ODEs: For $y^{(n)} = F(t, y, y', ..., y^{(n-1)})$. $\Rightarrow$ Let $x_i = y^{(i-1)}$. Then $x_n' = F(t, x_1, x_2, ..., x_{n-1})$ which is 1st ODE.

Def of Linear 1st ODEs: $\mathbf{y}'(t) = P(t)\mathbf{y}(t) + \mathbf{g}(t)$ $(\star)$, P is matrix-valued function. **Homogeneous**: $\mathbf{g}(t) = \mathbf{0}$.

Existence & Uniqueness Theorem: If P_{ij}, $\mathbf{g}$ of $(\star)$ continuous in R, $\mathbf{y}(t_0) = \mathbf{y}_0$ $\Rightarrow$ $\exists$ unique solution **y** in R' $R' = [t_0 - \varepsilon, t_0 + \varepsilon] \subseteq R = [t_0 - \delta, t_0 + \delta]$

2.1 Homogeneous

Solve: $\mathbf{y}' = P\mathbf{y}$ || If P constant, $P = A$ || If possible: λ, ξ, η be eigenvalues, eigenvectors, generalised eigenvectors.

Principle of Superposition: If $\mathbf{y}_i$ are solutions $\Rightarrow \sum_i c_i \mathbf{y}_i$ also. $S := \{\mathbf{y} | \mathbf{y}' = P\mathbf{y}\}$ is a *linear space*, $dim(S) = n$ if P_{ij} continuous.

Wronskian at x: $W[\mathbf{y}^{(1)}, ..., \mathbf{y}^{(n)}](t) = \begin{vmatrix} y_1^{(1)}(t) & \cdots & y_1^{(n)}(t) \\ \cdots & \cdots & \cdots \\ y_n^{(1)}(t) & \cdots & y_n^{(n)}(t) \end{vmatrix}$. $W(t) \neq 0, \forall t \Rightarrow \{\mathbf{y}^{(j)}\}$ **linearly independent.** ps: 经常用 Abel Theorem

· Geoemtric Meaning: Geometrically, W is the volume spanned by the three vectors $y^{(1)}, y^{(2)}, y^{(3)}$; t 增加或减小代表在 $y^{(i)}$ 方向上 compress/stretch.

Abel's Theorem: If $W(t_0) \neq 0, t_0 \in I$ $\Rightarrow$ $\forall t \in I, W(t) \neq 0$ || **Abel's Formula**: $W(t) = W(t_0)e^{\int_{t_0}^t tr[P(s)]ds}$; $t_0, t \in I$ $P(t)$ is continuous in I.

Fundamental Set of Solutions (FSoS): If $\{\mathbf{y}^{(j)}(t)\}$ *linearly independent* at $t \in I$, it is FSoS in I.

General Solution: If $\{\mathbf{y}^{(j)}(t)\}_{j=1}^n$ is FSoS (Linearly Independent) $\Rightarrow$ General solution: $\mathbf{y} = \sum_{j=1}^n c_j \mathbf{y}_j$

2.1.1 Eigenvalue/vector Method (A constant)

AM=GM|Real λ: general solution: $\mathbf{y} = \sum_j c_j e^{\lambda_j t} \xi_j$.

AM=GM|Complex λ: (ctd.) For $\lambda, \bar{\lambda} \in \mathbb{C}$. $\Rightarrow$ $e^{\lambda t}\xi = \mathbf{u} + \mathbf{v}i$. $\Rightarrow$ Independent sol: $\mathbf{u}, \mathbf{v}$ $\Rightarrow$ General solution: $\mathbf{y} = c_1\mathbf{u} + c_2\mathbf{v} + \cdots$

AM>GM: (ctd.) (For each ξ_λ, under λ) $\mathbf{y}^{(1)} = e^{\lambda t}\xi_\lambda$ $\mathbf{y}^{(2)} = e^{\lambda t}(t\xi_\lambda + \eta_1)$ $\mathbf{y}^{(3)} = e^{\lambda t}(\frac{t^2}{2!}\xi_\lambda + t\eta_2 + \eta_1)$, ... , $\mathbf{y}^{(n)} = e^{\lambda t}[\frac{t^{n-1}}{(n-1)!}\xi_\lambda + \sum_{i=0}^{n-2} \frac{t^i}{(i)!}\eta_{i+1}]$

2.1.2 Matrix Method | Exponential

Matrix Method: $\mathbf{y}(t) = e^{At}\mathbf{y}(0)$ (分两种方法: 1. 直接展开; 2. 通过一些已有 property 计算. ps: $e^{At} = \Psi$)

· **Property|AM=GM**: $e^{At} = Te^{Dt}T^{-1}$, $T = [\xi_1, ..., \xi_n]$, $D = diag(\lambda_1, ..., \lambda_n)$, $e^{Dt} = diag(e^{\lambda_1 t}, ..., e^{\lambda_n t})$

· **Property|AM>GM**: $e^{At} = Te^{Jt}T^{-1}$, $T = [\lambda_1, ..., \lambda_i, \eta_1^{(i)}, ..., \lambda_j, \eta_1^{(j)}, ...]$, $J = T^{-1}AT$ (Jordan form) , $e^{J_i t} = e^{\lambda_i t} \begin{pmatrix} 1 & t & \frac{t^2}{2} & \cdots & \frac{t^{n-1}}{(n-1)!} \\ 0 & 1 & t & \cdots & \frac{t^{n-2}}{(n-2)!} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$

Diagonalisation Method: (Ctd.) Let $\mathbf{y} = T\mathbf{x}$ $\Rightarrow$ $\mathbf{x}' = J\mathbf{x}$ [or $\mathbf{x}' = D\mathbf{x}$] 按前面方法解

Exponential of a matrix: $e^{At} := \sum_{n=0}^\infty \frac{(At)^n}{n!} = \mathbb{I} + At + \frac{1}{2!}A^2t^2 + \cdots = \lim_{n\to\infty}(\mathbb{I} + \frac{At}{n})^n$

· **Property**: $e^{A+B} = e^A e^B$ if $AB = BA$ || $e^{A \cdot 0} = \mathbb{I}$ || $[e^{At}]' = Ae^{At}$

2.1.3 Matrix Method | Fundamental Matrix

Fundamental matrix Ψ: If $\{\mathbf{y}^{(j)}(t)\}$ is FSoS. $\Rightarrow$ $\Psi(t) = [\mathbf{y}^{(1)}(t), ..., \mathbf{y}^{(n)}(t)]$.

· **Property**: $\Psi'(t) = P(t)\Psi(t)$ (It's a sol). || $det[\Psi(t)] = W(t) \neq 0$ || general solution: $\mathbf{y} = \Psi\mathbf{c}$, $\mathbf{c} = \Psi^{-1}(t_0)\mathbf{y}_0$

· **Relation with e^{At}**: $e^{At} = \Psi(t)$ with $\Psi(0) = \mathbb{I}$ || $e^{At} = \Psi(t)\Psi^{-1}(0)$

2.2 non-Homogeneous

Theorem: Solution of $(\star)$: $\mathbf{y} = \sum_i c_i \mathbf{y}^{(i)} + \mathbf{y}_p$, where $\mathbf{y}^{(i)}$ is solution of it's Homogeneous. ⇓通常会让特解中积分后的常数 c=0

Variation of Parameters: $\mathbf{y} = \mathbf{y}_g + \mathbf{y}_p$; $\mathbf{y}_g = \Psi(t)\Psi^{-1}(t_0)\mathbf{y}(t_0)$; $\mathbf{y}_p = \Psi(t)\mathbf{u}(t), \mathbf{u}(t) = \int_{t_0}^{t} \Psi^{-1}(s)\mathbf{g}(s)ds$. 经常 $\Psi(t)\mathbf{u}' = \mathbf{g}$ 来找 $\mathbf{u}$

Matrix methods|Diagonalisation: ^{1}Let $\mathbf{y} = T\mathbf{x}$ ^{2}$(\star) \Rightarrow \mathbf{x}' = D\mathbf{x} + T^{-1}\mathbf{g} = D\mathbf{x} + \mathbf{h}$ ^{3} 直接计算; $z_i = c_i e^{r_i t} + \underbrace{e^{r_i t}\int_{t_0}^{t} e^{-r_i s} h_i(s)ds}_{\mathbf{y}_p}$

Undetermined Coefficients: when $\mathbf{g}(t)$ is polynomials, or real/complex exponential. Let $P_n(t) = \sum_{i=0}^{n} \mathbf{p}_i t^i, A_n(t) = \sum_{i=0}^{n} \mathbf{a}_i t^i$

^{1} 构造 $\mathbf{y}_p$ 与 $\mathbf{g}$ 结构相匹配.(e.g. If $\mathbf{g} = P_n(t) \Rightarrow \mathbf{y}_p = A_n(t)$; If $\mathbf{g} = \mathbf{c}e^{kt} \Rightarrow \mathbf{y}_p = \mathbf{a}e^{kt}$) ^{2} 特殊: If $\mathbf{g} = \mathbf{c}e^{\lambda t}$,$\lambda$ is eigenvalue with $AM = n \Rightarrow \mathbf{y}_p = e^{\lambda t}\sum_{i=0}^{n} \mathbf{a}_i t^i$ ^{3} 带入原方程得未知系数

2.3 Geometrical Interpretation & Linear Approximation

* 此节中，Consider nonlinear autonomous system: $x' = F(x,y), y' = G(x,y)$ $(**)$

Phase Plane: O_{xy} referred as phase plane. **Trajectories**: Sol of ODE. $\mathbf{x}(t) = (x(t), y(t))^T$ direction $(dx/dt, dy/dt)$ **Phases Portrait**: 一组有代表性的轨迹

Critical Point (Equilibrium Point): $\mathbf{x}_0 = (x_0, y_0)$ is critical point iff $[x' = F(\mathbf{x}_0) = 0, y' = G(\mathbf{x}_0) = 0]$ or $[\mathbf{x}' = \mathbf{0}]$ (critical point 外变化简单)

Stable: Critical Point $\mathbf{x}_0$ is stable iff $\forall\varepsilon > 0, \exists\delta > 0$ s.t. $\forall$ sols $\mathbf{x}(t)$ with $||\mathbf{x}(0) - \mathbf{x}_0|| < \delta$, then $||\mathbf{x}(t) - \mathbf{x}_0|| < \varepsilon, \forall t \geq 0$

· 直观理解: 一开始离 critical point 很近, 那么随着时间推移也会保持在 critical point 附近. ps:attrcting 是会趋近于 critical point.

Attracting: Critical point $\mathbf{x}_0$ is attracting iff $\exists\delta > 0$ s.t. $\forall$ sols $\mathbf{x}(t)$ with $||\mathbf{x}(0) - \mathbf{x}_0|| < \delta$ satisfies: $\lim_{t\to\infty} \mathbf{x}(t) = \mathbf{x}_0$

Asymptotically Stable: Critical point $\mathbf{x}_0$ is asy.stab iff ^{1} Stable, ^{2} Attracting.

Theorem: If $\mathbf{x}_*(t) = (x_*, y_*)$ s.t. $F(\mathbf{x}_*), G(\mathbf{x}_*) \neq 0.$ $\Rightarrow$ $\exists H$ s.t. $H(x,y) = (\tilde{x}, \tilde{y}), \tilde{x}' = 1, \tilde{y}' = 0$ H is a smooth change of variable function in neighbourhood of $\mathbf{x}_*$

Non-Linear → Linear: For system $(**)$, assume $\mathbf{x}_0 = (x_0, y_0)$ is critical point. $\Rightarrow$ $\mathbf{u} = \mathbf{x} - \mathbf{x}_0$, then $\mathbf{u}' \approx A\mathbf{u}$ Linear; $A = \begin{pmatrix} \partial_x F(\mathbf{x}_0) & \partial_y F(\mathbf{x}_0) \\ \partial_x G(\mathbf{x}_0) & \partial_y G(\mathbf{x}_0) \end{pmatrix}$ Jacobian

2.4 Classification of Critical Points

* 此节中，Consider linear system: $\mathbf{x}' = A\mathbf{x}$. Critical Point(s): when $\mathbf{x}' = A\mathbf{x} = \mathbf{0}$

Classification of Critical Points at (x_0, y_0) (Linear): Let r_1, r_2 be sol of $det(A - \lambda I) = 0$. · $\mathbb{C}: r = \lambda \pm i\mu(\mu > 0)$

· If A constant, write sol: $\mathbf{x} = c_1 e^{r_1 t}\xi_1 + c_2 e^{r_2 t}\xi_2$ $\cdot GM = 1$: $\mathbf{x} = c_1 e^{rt}\xi + c_2 e^{rt}(t\xi + \eta)$ · AL: Almost Linear Systems.

$\mathbb{R}/\mathbb{C}$	Condition || Stability	Type || Name	Phase Plane Description	Other
$\mathbb{R}$	$r_1 < r_2 < 0$ || asy.stab	N || NSk	向原点,ξ_2 直线,ξ_1 曲线,and ξ_1 周围 $y = \pm x^3$	$c_2 \neq 0, t \to \infty$:ξ_2 主导方向; $c_2 = 0, t \to \infty$:ξ_1 主导方向
	$r_1 > r_2 > 0$ || unstable	N || NSo	原点向外,ξ_2 直线,ξ_1 曲线,and ξ_1 周围 $y = \pm x^3$	$c_1 \neq 0, t \to \infty$:ξ_1 主导方向; $c_1 = 0, t \to \infty$:ξ_2 主导方向
	$r_1 > 0 > r_2$ || unstable	SP || SP	$t \to \infty$,ξ_1 从原点向外,ξ_2 从外向原点 and: 像 $y = \pm\frac{1}{x}$, 同进同出	$t \to \pm\infty$: $\|\mathbf{x}\| \to \infty$; $t \to \infty$: $c_1, c_2 \neq 0, \|\mathbf{x}\| \to \infty, \xi_1$ 主导; $t \to \infty$: $c_2 = 0, \|\mathbf{x}\| \to \infty, \xi_1$ 主导; $t \to \infty$: $c_1 = 0, \|\mathbf{x}\| \to 0, \xi_2$ 主导
	$r_1 = r_2 < 0$, GM=2 || asy.stab	PN || PN or Stable Star	直线 向原点	直线, u_1/u_2 is t independent
	$r_1 = r_2 > 0$, GM=2 || unstable	PN || PN or Unstable Star	直线 从原点向外	直线, u_1/u_2 is t independent
	$r_1 = r_2 < 0$, GM=1 || asy.stab	IN (AL:Type: SpP) || IN (Stable)	S 曲线, 向原点	$t \to \infty, \|\mathbf{x}\| \to 0, \xi$ 主导 ps: 旋转方向大体和 $\eta + c_2\xi$ 方向相同
	$r_1 = r_2 > 0$, GM=1 || unstable	IN (AL:Type: SpP) || IN (Unstable)	S 曲线, 从原点向外	$t \to \infty, \|\mathbf{x}\| \to \infty, \xi$ 主导 ps: 旋转方向大体和 $\eta + c_2\xi$ 方向相同
$\mathbb{C}$	$\lambda \neq 0, \lambda > 0$ || unstable	SpP || Unstable Focus	向外椭圆 (elliptical) 螺旋	$t \to \infty, \|\mathbf{x}\| \to \infty$ ps: 考虑 $A = (a,b;c,d)$, 如果 bc>0, 顺时针，如果 bc<0, 逆时针
	$\lambda \neq 0, \lambda < 0$ || asy.stab	SpP || Stable Focus	向内椭圆 (elliptical) 螺旋	$t \to \infty, \|\mathbf{x}\| \to 0$ ps: 考虑 $A = (a,b;c,d)$, 如果 bc>0, 顺时针，如果 bc<0, 逆时针
	$\lambda = 0$ || stable (AL:Indeterminate)	C (AL:C or SpP) || C	椭圆 (elliptical) and 半长轴 ξ 实部方向	Bounded trajectory or $\exists$ Periodic Trajectories
ps: N = Node | PN = Proper Node | IN = Improper / Degenerate Node | SP = Saddle Point | SpP = spiral (focus) point | C = Center | NSk = Nodal Sink | NSo = Nodal Source				
For Almost Linear systems (见 2.5): $\mathbf{u}' = A\mathbf{u} + \underline{\eta}$, 如果 linear Approximation (去掉 $\underline{\eta}$), 按照 Linear 考虑. 如果不去 $\underline{\eta}$, 按照去掉的算 r_1, r_2, Type 按照正常 / (AL) 的来.(图像会到 $(0,0)$)				

Some Defs: $Separatrix$: 将不同区域分开的轨迹; $Basin\ of\ attraction$: 所有趋近 critical point 的轨迹. ps: In any nonlinear ODEs, it's important to determine the *basin of attraction*.

3 Non-Linear ODE Methods

* 此节中，Consider two-dimensional autonomous system: $x' = F(x,y), y' = G(x,y)$ and $\mathbf{x} = (x,y)^T$ $(\#)$ Lyapunov's: 不需要知道微分方程的确切解 critical value. Poincaré-Bendixson: 寻找 limit cycle

Positive (Negative) [semi-]definite: Function $E: D \subset \mathbb{R}^2 \to \mathbb{R}$ is positive (negative) [semi-]definite iff: ps: positive [semi-]definite 正 (半) 定

$$^1E(\mathbf{x}_0) = 0, {}^2\forall\mathbf{x} \in D, \mathbf{x} \neq \mathbf{x}_0, E(x,y) > 0 \text{ or } (E(x,y) < 0) \text{ or } [E(x,y) \geq 0]$$ ps: Neighbourhood 领域

Isolated Critical Point: Critical Point $\mathbf{x}_0$ *isolated* iff $\exists$ Neighbourhood containing no other critical points. ↓⇒ E :Lyapunov function

Lyapunov Stability Theorem: ^{1}(#); ^{2}ICP $\mathbf{x}_0 \in D$; $^3\exists$ 正定且连续 $E: D \subset \mathbb{R}^2 \to \mathbb{R}, E(\mathbf{x}_0) = 0$; || $^4\frac{dE}{dt} = \nabla E \cdot \mathbf{x}' = \partial_x E \cdot F + \partial_y E \cdot G$ 负 (半) 定 on $D \setminus \mathbf{x}_0(D) \Rightarrow \mathbf{x}_0$ asp.(stab)

· **Look for function E**: Way 1: May be energy, but not always. Way 2: *e.g.* $E = ax^{2n} + by^{2m}$,$E = a(x+y)^{2n} + by^{2m}$ ps: $\frac{dE}{dt} = \nabla E \cdot \mathbf{x}'$

Lyapunov Instability Theorem: ^{1}(#); ^{2}ICP $\mathbf{x}_0 \in D$; $^3\exists E: D \subset \mathbb{R}^2 \to \mathbb{R}, \partial_i E$ 连续,$E(\mathbf{x}_0) = 0$; $^4\forall$ 领域$\mathbf{x}_0, \exists\mathbf{x}_1$ $E(\mathbf{x}_1) > 0$; || $^5\frac{dE}{dt}$ 正定 on $D \Rightarrow \mathbf{x}_0$ unstable

Periodic Solution: Soln $\mathbf{x}$ has: $\mathbf{x}(t+T) = \mathbf{x}(t), \forall x$. || 闭合曲线 ⇔ Periodic sol || Periodic Solution + 附近有吸引或排斥 ⇒ Limit Cycle || when $\dot{r} = 0, \dot{\theta} = const$. Periodic sol

Limit cycle: Limit cycle ⇒ Periodic Solution || Def $t \to \pm\infty$, $\exists$ non-closed trajectory asymptotes to it. (inside or outside) ps: It's Orbital Stability

1. Limit cycle is: ^{1}asy.stab: $\forall$ trajectories asy. it; ^{2}semistable: $\exists$ trajectories 单方向进/出; ^{3}unstable: $\forall$ trajectories away from it.
2. **Theorem (If $\exists$ limit cycle / closed)**: $^1F, G$ 一阶偏导连续 in simply connected D; $^2\exists$ 极限环/封闭环 $(\star)$ in D. $\Rightarrow$ $^1\exists$ CP in $(\star)$; ^{2}If $\exists!$ CP in $(\star)$, it's not SP.
3. **Theorem (No limit cycle / closed)**: $^1F, G$ 一阶偏导连续 in simply connected D; $^2\partial_x F + \partial_y G$ 符号不变 on D $\Rightarrow$ No closed trajectory in D.

   2 的推论: ^{1}If simple connected D no CP $\Rightarrow$ No closed trajectory in D. ^{2}If simple connected D has unique CP $\Rightarrow$ No closed trajectory in D

Poincaré-Bendixson Theorem: $^1F, G$ 一阶偏导连续 in D; $^2\exists D_1 \subset D$ bounded; $^3R = D_1 \cup \partial D_1$ has NO CP; || $^4\exists$ soln $\mathbf{x} \in R, \forall t \geq t_0$ $\Rightarrow$

$\Rightarrow$ $^1\exists$ closed trajectory. ^{2} Either $\mathbf{x}$ *sprial towards a Periodic trajectory.* or $\mathbf{x}$ *periodic* ^3R is not simply connected

· **Find Trapping Region**: **trapping region**: 符合上面定理 2-4 除 CP 的 region. Method: Consider *polar coordinate*, 计算 $r^2 = x^2 + y^2$ 对 t 的求导, 寻找 $r\dot{r} \geq 0$ or $r\dot{r} \leq 0$

4 Fourier Series

* BVPs 可能没有解/一个解/无数解; IVP 只会有一个解

4.1 Function Space, Eigenfunctions, Periodic Function and Orthogonal Function

Periodic Functions: Function $f(x)$ is Periodic iff $f(x+T)=f(x), \forall x$. **Period**: T. **Fundamental Period**: The smallest possible T.

· Property: If f, g has same period T. $\Rightarrow$ $af(x)+bg(x)$ also has period T. || For $sin(\omega x), cos(\omega x)$: $T = 2\pi/\omega$

Differential Operators: Any linear combination of n-th derivatives defines a linear map or linear operator. e.g. $p(x)\frac{d}{dx}+\frac{d^2}{dx^2}: y \mapsto p(x)\frac{dy}{dx}+\frac{d^2y}{dx^2}$

Eigenvalues & Eigenfunctions: If D is a *differential operator* $\Rightarrow$ If $Df=\lambda f$, then λ is **Eigenvalue** and f is **Eigenfunctions**

Inner Product: $(f,g)=\int_a^b f(x)\overline{g(x)}dx$ For funcs f, g in (a,b) || **Orthogonal**: $(f,g)=0$ || **Square Norm**: (f,f).

Kronecker delta: $\delta_{mn} = \{0, m \neq n;\ 1, m = n.$

4.2 Fourier Series

Fourier Series: Periodic function with $T=2L$, $f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}\left(a_n cos(\frac{n\pi x}{L})+b_n sin(\frac{n\pi x}{L})\right)$ (*) where a_0, a_n, b_n are Fourier Coefficients.

Euler-Fourier Formulas: Suppose (*) converges in the interval $[-L, L]$. Then:

$$a_n=\frac{1}{L}\int_{-L}^{L} f(x)cos(\frac{n\pi x}{L})dx \qquad b_n=\frac{1}{L}\int_{-L}^{L} f(x)sin(\frac{n\pi x}{L})dx \qquad a_0=\frac{1}{L}\int_{-L}^{L} f(x)dx$$ (Average Value of f over $2L$)

Piecewise Continuous: $f(x)$ piecewise continuous on $[a,b]$ iff $\exists a=x_0<\cdots<x_n=b$ s.t. [1] $f(x)$ continuous $\forall x \in (x_{i-1}, x_i)$ [2] $f(x_i^{\pm})=\lim_{x\to x_i^{\pm}} f(x)$ finite.

Fourier Convergence Theorem: If [1] f, f' piecewise continuous on $[-L, L]$, [2] $f(x+2L)=f(x)$. $\Rightarrow$ Fourier Series of f converges.

· Futhermore, Let Fourier Series of f is F. $F(x)\to f(x)$, $f(x)$ continuous; $F(x)\to\frac{f(x^+)+f(x^-)}{2}$, $f(x)$ discontinuous.

Theorem: Fourier Series for piecewise continuous functions can be Integrated term-by-term to Convergent Fourier Seires.

Finite Truncation (Partial Sum): Finite Truncation of Fourier Series is: $s_k=\frac{a_0}{2}+\sum_{n=1}^{k}\left(a_n cos(\frac{n\pi x}{L})+b_n sin(\frac{n\pi x}{L})\right)$ || **Error**: $e_N = s_N - f(x)$

· $n\to\infty$: [1] Continuous points, Fourier approximation of $f(x)$ is better ($|e_n|$ decrease). [2] Discontinuous points, oscillation of Fourier series get smaller.

· **Gibbs Phenomenon**: The Amplitude of the *largest* oscillation (near discontinuous points) does *not decrease*.

4.3 Other Results on Fourier Series

Extension of a function: $f(x)$ know for $x\in[0,L]$ $\Rightarrow$ Even Ext.: $h(x)=\{f(x),[0,L];\ f(-x),(-L,0)$ || Odd Ext.: $h(x)=\{f(x),(0,L);\ 0, x=\pm L, 0;\ -f(-x),(-L,0)$

Complex Form of Fourier Series: $f(x)=\sum_{-\infty}^{\infty} c_n e^{\frac{i\,n\pi x}{L}}$ || $c_n=\frac{a_n-ib_n}{2}; c_{-n}=\frac{a_n+ib_n}{2}; c_0=\frac{a_0}{2}, n>0$ || $a_0=2c_0; a_n=c_n+c_{-n}; b_n=i(c_n-c_{-n})$

Euler-Fourier Formula (Complex): If (⇑) of Periodic f, $(T=2L)$ converges in $[-L, L]$ $\Rightarrow$ $c_n=\frac{1}{2L}\int_{-L}^{L} f(x)e^{-\frac{i\,n\pi x}{L}}$ || $c_{-n}=\overline{c_n}$ if f real.

Parseval's Theorem: If [1] Periodic function f; [2] f has convergent Fourier Series.

$\Rightarrow$ Square Norm $(f,f)=\int_{-L}^{L}|f(x)|^2dx=2L\sum_{-\infty}^{\infty}|c_n|^2=L\left[\frac{|a_0|^2}{2}+\sum_{n=1}^{\infty}(|a_n|^2+|b_n|^2)\right]$ ps: 无限维的勾股定理

5 Appendix

Trigonometric func: s+s=sc,s-s=cs,c+c=cc,c-c=-ss || e.g. $sinA+sinB=2sin(\frac{A+B}{2})cos(\frac{A+B}{2})$ $sinAsinB=-\frac{1}{2}[cos(A+B)-cos(A-B)]$

hyperbolic func: $sinh(x)=\frac{e^x-e^{-x}}{2}$ $cosh(x)=\frac{e^x+e^{-x}}{2}$ $cosh^2(x)-sinh^2(x)=1$ $[sinh(x)]'=cosh(x)$ $[cosh(x)]'=sinh(x)$

Common Int: $\int tan(x)dx=-ln|cos(x)|$; $\int cot(x)dx=ln|sin(x)|$; $\int sec(x)dx=ln|sec(x)+tan(x)|$; $\int csc(x)dx=ln|csc(x)-cot(x)|=ln|tan(\frac{x}{2})|$

Common Der: $[tan(x)]'=sec^2(x)$; $[cot(x)]'=-csc^2(x)$; $[sec(x)]'=sec(x)tan(x)$; $[csc(x)]'=-csc(x)cot(x)$; $[arctan(x)]'=\frac{1}{1+x^2}$.

· $1+tan^2(x)=sec^2(x)$; $1-csc^2(x)=cot^2(x)$; $csc^2(x)-cot^2(x)=1$. $[arcsin(x)]'=\frac{1}{\sqrt{1-x^2}}$; $[arccos(x)]'=-\frac{1}{\sqrt{1-x^2}}$

Polar Coordinate: $x=rcos(\theta)$,$y=rsin(\theta)$ we have: $r\dot{r}=x\dot{x}+y\dot{y}$, $y\dot{x}-x\dot{y}=-r^2\dot{\theta}$

Explain of Fourier Series: In application, it's decomposition of signal into infinite (countable) waves; each wave $cos(\frac{n\pi x}{L}), cos(\frac{m\pi x}{L})$ has discrete period; a_n, b_m correspond to amplitudes of waves. || In vector algebra, it's a space of periodic functions; f corresponds to an arbitrary vector **v**; orthogonal basis $\{1, cos(\frac{n\pi x}{L}), sin(\frac{m\pi x}{L})\}$; a_n, b_m correspond to coefficient of each components of basis.

Even & Odd Function: Even: $f(-x)=f(x)$; Odd: $f(-x)=-f(x)$. || Property: [1] $f(x)$ odd: $\int_{-L}^{L} f(x)dx=0$ [2] $f(x)$ even: $\int_{-L}^{L} f(x)dx=2\int_0^L f(x)dx$

· Other Property: [1] even + even = even; [2] odd + odd = odd; [3] even × even = even; [4] odd × odd = even; [5] even × odd = odd.